Pact: Verified Settlement for Agent-to-Agent Commerce

The acceptance checks are written by the negotiation itself, and every deal carries a seller bond that is paid to the buyer when the checks fail

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Abstract: *AI agents have started buying services from each other. The payment rails for that market are shipping fast. The trust layer is not. When one agent pays another for a piece of work, nothing in the current stack reads the delivery against what the two sides agreed before the money settles. A seller who delivers junk loses nothing but time. This paper describes Pact, a system I built to close both gaps. Two agents first negotiate a price. Inside the same negotiation, they also write the acceptance checks for the deal, trading values back and forth until both sides accept the set. The buyer then pays through PayPal, and the seller posts a bond worth 20 percent of the price through a second PayPal order. A separate verifier runs the frozen checks against the delivered work. If every check passes, the seller is paid and the bond comes back. If some checks pass, the deal settles graded: the seller is paid for the share that passed, the buyer is refunded the rest, and the bond is split in the same proportion. If no check passes, the buyer is refunded in full and the whole bond goes to the buyer. I ran the full loop on the real PayPal sandbox and tested each part on its own. A hand-built battery of 275 verifier checks produced no errors. A Monte Carlo run of 10,000 simulated settlements created and lost no money. Four live sandbox deals settled on October 22, 2026, and when I read all 16 of their money movements back from PayPal on November 12, 2026, every one matched the record written on the day.*

Index Terms: *agent commerce, escrow, PayPal, seller bond, settlement verification*

I. INTRODUCTION

Picture a simple job. A buyer agent needs a logo and brand kit. A seller agent offers to make one. They agree on a price in seconds. Then the hard part starts. Who holds the money while the work is done? Who reads the delivery and decides whether it is what was promised? What happens to a seller who sends back recycled filler and hopes nobody looks?

Humans have run this exact market for years, and the record is not good. In a 2015 Freelancers Union survey, 71 percent of freelancers said they had trouble getting paid at some point in their career, and the average loss came to more than $6,000 a year [1]. The 2019 Freelancing in America study by Upwork and Freelancers Union found that 74 percent of freelancers had faced late payment or no payment at all, and only 28 percent consistently worked under a written contract [2]. The fix that exists is priced for large claims. On Upwork, a binding arbitration for a claim under $20,000 costs $675 in total, split between the two sides [3]. On a $150 logo job, the decision costs more than four times the job.

Paying directly does not close the gap either. PayPal Purchase Protection is built around physical goods: an item that never arrives, or an item that is clearly not what was described. PayPal's own page lists services and intangible items as not covered [8]. A freelance service paid with a plain PayPal payment has no escrow, no milestone, and no reading of the delivery against any agreement.

The agent payment systems that shipped in 2025 solve a different problem. They prove that an agent was allowed to spend, and they move the money. Section II covers them in detail. None of them reads a delivery against negotiated acceptance checks. None of them makes the seller post a bond. And instant, final settlement removes the refund path that a bad delivery needs.

Pact is my attempt at the missing layer. It rests on three design choices.

First, the contract is written by the negotiation itself. The buyer proposes a value for each acceptance check. The seller accepts or counters. The settled values freeze, and the seller is later judged only on terms it agreed to.

Second, every deal carries its own seller bond. A refund alone hands the buyer his money back and costs the seller nothing. A bond that can be cut changes the arithmetic of cheating. Failure stops being free.

Third, settlement is graded and runs on PayPal rails. The seller is paid for the checks that passed, not all or nothing. Orders, payouts, and capture refunds are real PayPal calls, and the account holder approval step that PayPal requires stays in the open instead of being worked around.

The rest of the paper is organized as follows. Section II places Pact next to escrow services, buyer protection, and the agent payment protocols. Section III describes the system. Section IV reports the tests: verifier accuracy, a labeled benchmark, negotiation and settlement sweeps, a Monte Carlo run, property tests, an adversarial battery, four live sandbox deals

with a later read-back, and 29 cases priced from real marketplace listings. Section V works through what one deal earns the platform that runs it. Section VI states the limits plainly. Section VII concludes.

II. RELATED WORK

A. Escrow and Arbitration on Freelance Platforms

Upwork's fixed-price protection is the strongest form of the current model. The client funds an escrow account before work starts. If the two sides disagree, staff mediation comes first. If mediation fails, a third-party company runs binding arbitration [3]. The money holding works. The judging is the problem for small jobs: a human reads the file, the process takes time, and the $675 fee for claims under $20,000 is larger than most single deliverables are worth. Escrow answers the question of who holds the money. It does not answer who reads the work, at a price a small job can carry.

B. Buyer Protection on Payment Networks

PayPal Purchase Protection covers two cases for eligible purchases: the item did not arrive, or the item is significantly not as described. Services and intangible goods sit outside the program [8]. That boundary made sense when the buyer was a person shopping for objects. It leaves agent-bought services, which are intangible by definition, with no protection at all when they are paid for directly.

C. Agent Payment Protocols

Three protocols from 2025 define the current state of agent payments. Google's Agent Payments Protocol, AP2, uses signed mandates to prove that a user authorized an agent to act and pay [9]. The Agentic Commerce Protocol from OpenAI and Stripe lets an agent complete a purchase inside a conversation, with the merchant's existing payment stack behind it [10]. Coinbase's x402 protocol revives the HTTP 402 status code so a machine can pay for a request at the moment it makes one [11]. All three answer the same two questions: may this agent spend, and how does the money move. None of them reads what was delivered. None of them holds money back against a verdict. Their settlement is designed to be fast and final, which is the right call for buying a known item and the wrong call for paying a stranger for work that does not exist yet.

D. Escrow Prototypes for Agents

Escrow is an old idea, and builders have started pairing it with automated judging in hackathon and research prototypes. In the public projects I reviewed, the pieces appear one or two at a time: an escrow wallet here, a model acting as judge there, a fixed checklist from the platform in a third. I did not find a system where the haggling between the two agents writes the acceptance checks, where the size of the seller bond is itself set inside that same negotiation, and where the money then settles on fiat payment rails with a working refund path. That combination, stated narrowly, is what this paper claims.

III. SYSTEM DESIGN

A. One Deal, End to End

A deal moves through nine steps.

1. A buyer agent and a seller agent negotiate the price.
2. The two agents write the contract: values for each acceptance check, proposed and countered until they settle.
3. The buyer pays the agreed price through a PayPal order. The seller posts the bond, 20 percent of the price, through a second PayPal order. Both orders are captured only after the account holders approve them.
4. The seller delivers the work inside the app.
5. A verifier agent runs every frozen check against the delivery and records a verdict per check.
6. Full pass: the price is paid out to the seller and the bond is returned.
7. Partial pass: the seller gets one revision round. A delivery that passes no check skips revision and settles at once.
8. Final partial verdict: the deal settles graded, as Section III-F describes. Final zero verdict: full refund to the buyer, full bond paid to the buyer.
9. Every event, from the first negotiation move to the last payout, lands in an audit ledger.

B. Negotiation

On price, each side gives up a share of the remaining gap on its own turn. Every move carries a short written reason. The round cap is enforced: talks that run past it end with no deal. If the seller's floor sits above the buyer's ceiling, the deal walks away and no money moves.

A deal can also run in multi-issue mode. After the price locks, the agents trade three more terms: the delivery window, the number of revision rounds, and the bond percent inside a band of 10 to 30. Each side holds simple utility weights, and the side to move gives ground where its concession buys the other side the most. In practice the buyer gives ground on the delivery window and the seller gives ground on revision rounds, while the bond lands near the middle of the band. The settled terms freeze into the contract, and the bond that gets charged is the negotiated percent, not a platform constant. Section IV-C reports that turning this phase on never changed whether a deal agreed.

C. Contract Writing

The buyer opens by proposing a value for each check in the scenario's check list. The seller accepts a value or counters it. When both sides have numbers on the table for the same check, the settled value is the midpoint. The set that results is

the contract. It freezes. Nothing in it can change after money moves.

The verifier speaks five check kinds: counted content items (for example, at least three logo concepts), hex colors, hashtags, word counts, and required strings. Five kinds sounds small. Section IV shows how much behavior they cover across fifteen scenarios, and Section VI is honest about what they cannot judge.

D. Money Movement and the Approval Checkpoint

Orders run through PayPal's official Agent Toolkit, the Python package paypal-agent-toolkit [6], [7]. Pact calls the toolkit's dispatch core for create_order, pay_order, and get_order_details. The toolkit ships no payout tool and no refund tool, so payouts and capture refunds run on a small direct REST client against the PayPal API [4], [5].

My first design assumed code could capture an order right after creating it. PayPal refuses: the capture call fails with ORDER_NOT_APPROVED, because the account holder must approve the order in PayPal's own checkout first. I rebuilt the payment step around that fact instead of against it. The agents prepare both orders. The deal parks in an awaiting-approval state with one approval link per order. Capture runs only after both approvals land, and each capture is guarded so a retry after a partial approval is safe. I kept this checkpoint on purpose. It matches how PayPal itself frames agentic payments: agents propose and prepare, people approve money movement. Both approvals are ledger events.

One honest note on the hold. In this build, both captures land in the platform's own account, so the platform holds both sides' money and carries both capture fees. The held state in the ledger models PayPal's delayed disbursement, the native hold primitive of the Commerce Platform. The production shape, described in Section V, makes the seller the payee, disburses on a verification pass, and takes the platform fee inside the disbursement instruction.

E. The Contract Travels with the Money

When the spec freezes, the app hashes it with SHA-256. The hash and the deal id are written onto both PayPal orders: into the purchase unit custom_id field on the REST path, and into the line item name on the toolkit path, because the toolkit's create_order exposes no custom_id field. The verification summary records the same hash. Anyone holding a PayPal order record can hash the frozen spec and compare. Change one check value and the hash changes. The contract that judged the delivery is provably the contract the money moved under.

F. Verification and Graded Settlement

The verifier is deterministic. For each check it stores the measured value, the verdict, and a plain reason. Money moves on the verifier's output, never on a model's say-so. An optional language model layer can rewrite the wording of negotiation reasons and verdict reasons so they read naturally. It decides nothing: prices, check values, verdicts, and payments all come from the deterministic engine, and the app runs fully with no model key set.

Settlement follows the verdict. A full pass releases the price and returns the bond. A partial pass earns the seller exactly one revision round. After the final verdict, a partial result settles graded. The split is computed in integer cents with a single half-even rounding step, so no cent appears or vanishes. A deal that passes 3 of 5 checks pays the seller three fifths of the price, refunds the buyer two fifths, returns three fifths of the bond, and cuts two fifths of it to the buyer. A zero pass skips revision, refunds the buyer in full, and pays the whole bond to the buyer.

Each money leg (price payout, bond return, price refund, bond slash) is recorded as it completes. If a leg fails midway, the deal stays resumable and a retry runs only the legs that are missing, each with a deterministic idempotency key so no leg can run twice. A per-deal lock wraps every state-changing call: parallel calls get one success and the rest get a 409. Every payout is read back after acceptance, and the deal stays pending until PayPal reports the payout successful. A denied payout writes a correcting ledger event and retries under a fresh key rather than recording a release that never happened.

G. Timeout

A seller can also fail by silence. If the delivery window passes with no delivery, the buyer may end the deal: full refund, full bond slash. On a single-issue deal the window is 7 days. A deal sitting in revision cannot be timed out, because the seller did deliver and the buyer is the one waiting on the second attempt.

H. Reputation

Each agent carries a reputation score that moves with outcomes, in plain arithmetic. A released deal adds 6. A revision round costs the seller 3. A refund costs 12. A slashed bond costs 20 more. A graded deal takes one mixed step: the passed share of the 6 point gain minus the failed share of the 20 point slash penalty. The cheat deal in Section IV-H dropped its seller from 64 to 46 on a single deal.

I. Ledger and Dispute Evidence

The audit ledger records every negotiation move, the frozen contract, each order, approval, capture, delivery, verdict, payout, refund, bond movement, and reputation change. The interface renders it as a filterable grid (AG Grid) with event families color coded and a quick filter across the whole history. From any settled deal, the app produces a dispute evidence packet: the frozen contract, its hash, the PayPal records, and the verdict trail, in one downloadable file. The

packet exists so that a dispute starts from evidence instead of from two agents arguing.

IV. EXPERIMENTS

Every number in this section comes from a run stored in the project repository [13]. Seeds are fixed where randomness is involved. Tests that do not need real money run in demo mode, where a local payment simulator stands in for PayPal and stamps every record as demo. The live runs in Section IV-H used the real PayPal sandbox.

A. Verifier Accuracy Battery

I built 55 deliveries by hand across the scenario set: honest work, weak work, and work designed to trip a specific check. The battery judged 275 checks. At check level it scored 207 true passes, 68 true fails, no false passes, and no false fails. Delivery level accuracy matched: every delivery got the overall verdict its construction called for. Four of the deliveries were trap cases built around known counter weaknesses, and all four were judged right. One trap matters enough to name here: a delivery whose text carried hex color codes but no hashtags. Section IV-G explains what it caught.

B. PactBench

PactBench is the standing regression set: 337 labeled deliveries, each judged on 5 checks, for 1,685 judged checks in total. Labels come from how each delivery was built (honest, weak, or cheating) across the fifteen scenarios, so the set can be regenerated and extended. Every verifier change runs against PactBench before it ships. I use the 275 check battery in Section IV-A for the accuracy claim, because its ground truth was set by hand per check. PactBench exists to catch regressions at scale, and it is reported here as a size and a role, not as an accuracy figure.

C. Negotiation Sweeps

The negotiation sweep ran 196 deals across four concession rates (0.2, 0.35, 0.45, and 0.6 of the remaining gap per move). In every run where a deal zone existed, the agents reached agreement: a rate of 1.0. In every run where the seller floor sat above the buyer ceiling, the agents walked away: also 1.0. Agreement took 2.79 price moves on average. The multi-issue mode ran its own sweep against matched single-issue deals, and the terms phase changed the agreement outcome in zero runs. Terms moved; the decision to deal did not.

D. Settlement Oracle

Ten end-to-end settlement paths (honest, weak, and cheat runs across five scenarios) were scored against a rules oracle written independently of the settlement code. For each path the oracle predicts the final state of the deal, the price, and the bond, plus the reputation step. All ten paths matched the oracle on every compared field.

E. Monte Carlo Settlement Run

The Monte Carlo run drove 10,000 randomized deals through the real settlement engine end to end. Two invariants were asserted on every deal. First, the terminal state must match the final verdict: a deal that passed everything never ends refunded with a cut bond, a deal that failed a check never ends released, a partial pass ends graded, and a zero pass ends refunded. Second, money must conserve per deal, in integer cents: payouts plus refunds plus returned bond plus cut bond plus platform fees equal the captured totals, and money out never exceeds money in. The run injected fail-once faults into settlement legs (1,268 price payouts, 1,269 bond returns, 1,093 price refunds, and 1,102 bond slashes were forced to fail once and then resume) to prove the resumability path under the same invariants. Result: zero violations of either invariant across all 10,000 deals. Of the deals, 4,591 ended released, 4,009 refunded, and 1,228 graded.

F. Property Tests

A randomized property suite ran 401 cases against the live API: full deals across scenarios and quality levels, illegal state transitions, and fuzzed malformed request bodies. The suite executes 5,285 individual checks. Illegal transitions were rejected, malformed bodies were refused with clean errors, and the state machine invariants held throughout.

G. Adversarial Battery

I attacked the system from both sides. The seller battery holds seven attacks: section markers with the content stripped out, word stuffing in three forms (a single repeated word, rotated sentences, and an honest front with a stuffed tail), a weak delivery padded with bare section words, honest work with one concept removed, and hashtag stuffing against the social pack spec. The API battery holds six attacks on the state machine, including verifying before delivery, paying twice on one deal, delivering after a refund, and re-verifying a passed deal to force a refund. After the fixes described below, no seller attack passed all five of its checks (the best any attack managed was four of five), and all six API attacks were blocked.

The battery earned its keep twice, and so did the traps in Section IV-A.

The hashtag bug. The first hashtag counter matched a plain token pattern. A hex color code such as #AABBCC matched it, so a delivery with two color codes and no hashtags passed a hashtag check it should have failed. The trap case caught it. The fix is a ruling inside the counter: a token that is exactly a six digit hex value counts as a color and never as a hashtag. The case stays in the battery as a regression test.

The item counting bug. The first item counter counted bare section markers. A delivery that listed its markers with every word of content stripped out passed counted-item checks. The

adversarial battery found this one; the happy path tests had not. The fix requires at least three words of content after a marker before the item counts.

The battery also probed deal creation. A zero percent bond and a negative price were both accepted. Both are rejected now. A zero bond makes cheating free, so it cannot be a valid input.

H. Live Sandbox Deals and the Read-Back

On October 22, 2026, the full loop ran on the real PayPal sandbox: toolkit order creation, account holder approval of both orders, capture, delivery, verification, and settlement, recorded with real sandbox transaction IDs.

The cheat deal is the one worth a slow read. The seller sent recycled copy that ignored the frozen contract. The verifier passed the one check the copy happened to satisfy, sent the deal into its single revision round, and caught the near identical second delivery again. The deal settled graded: the seller was paid $36.19 of the $180.96 price, the buyer was refunded $144.77, and $28.95 of the $36.19 bond was cut to the buyer. The seller's reputation fell from 64 to 46 on that one deal.

Three weeks later, on November 12, 2026, I read all 16 money legs behind these four deals back from the PayPal sandbox through the API: every capture, every payout batch and its item, and every refund. The read moved no money. The released deals show captures COMPLETED and payouts SUCCESS. The failed deals show the price capture REFUNDED, the refund COMPLETED, and the bond slash payout SUCCESS to the buyer account. Sixteen of sixteen legs match the outcome recorded when the deals settled.

I. Cases Priced from Real Listings

Beyond the sandbox deals, 29 cases were priced from live Fiverr and Upwork listings and run end to end. A listing at $74.06, a logo kit at $120.64, a coding task at $49.22, a research brief at $78.52, and a labeling batch at $54.00 are representative. One case buys a physical product: a procurement deal for Sony WH-1000XM5 headphones, grounded in a live Channel3 search that returned offers from $239.97 to $336.99 across five retailers, with a median of $298 [13]. The deal negotiated to $270.52 and released after all five checks passed. The cheat cases in the set settled the way the checks dictated. A social pack cheat at a $137.09 price settled graded: the seller was paid $27.42, the buyer was refunded $109.67, and $21.94 of the $27.42 bond was cut to the buyer.

V. ECONOMIC ANALYSIS

A trust layer is only worth building if the platform running it gets paid. This section works the arithmetic for PayPal under stated rates and assumptions. It is a model, not a measurement.

One Pact deal puts up to four money movements on PayPal rails instead of one: the price capture, the bond capture, and then either two payouts or a refund plus one payout. Take PayPal's published US rates: 3.49 percent plus $0.49 to receive a commercial payment, $0.25 per Payouts API payout, and capture fees kept when a payment is refunded. A completed $250 deal with a $50 bond earns about $11.95 in gross fees. A refunded deal earns about $11.70, because the capture fees were collected up front and are not returned. Failure costs the platform almost nothing. Measured against the $300 captured on the deal, the effective take is about 4.0 percent, next to a blended PayPal take rate of roughly 1.6 percent. At one million deals a month, the same rates give roughly $143 million a year in gross fees on $3.6 billion of new volume. On the production shape from Section III-D (seller as payee, delayed disbursement, platform fee inside the disbursement instruction, bond as authorize and void), the two payout fees disappear and the same volume gives about $115 million a year. The full working, with sources for each rate, sits in the project repository [13].

The strategic point is larger than the fee point. PayPal's chief executive Enrique Lores has argued in public that trust mattered when e-commerce started and will matter even more in an agentic world, and that holding that position is PayPal's job [12]. Verified settlement with a seller bond is that position expressed as a product: held money, a computed verdict instead of a support queue, refunds that actually happen, and a funded bond that makes the buyer whole out of the seller's stake instead of the platform's balance. PayPal absorbed about $1.3 billion in transaction losses in 2025 across fraud, chargebacks, and protection payouts. A layer that settles most disputes by arithmetic before they become claims works directly on that line.

VI. LIMITATIONS

The verifier judges shape, not craft. A word count proves length. A required string proves presence. Neither proves the words are good. A fluent delivery and a hollow one can pass the same five checks if both hit the agreed counts and terms. The bond and the other checks still bind, and reputation still moves, but nobody should read a pass as a quality award.

Padding is a live weakness. Repetition screens catch crude stuffing, including the single repeated word attack in Section IV-G. Padding written with varied, natural-sounding filler can still satisfy a length check, because a length check measures length. I left this in the open rather than tuning the screens until the battery turned green; the honest statement is that length is a weak signal and the system treats it as one signal among five.

The accuracy numbers measure rule correctness on deliveries I constructed. Real freelance work is messier than my honest, weak, and cheat builders. PactBench labels come from construction, so its 337 deliveries prove the engine follows its

TABLE I
LIVE SANDBOX DEALS, OCTOBER 22, 2026

DEAL	SCENARIO	PRICE	BOND	OUTCOME	PAYPAL RECORDS
deal-0001	Logo and brand kit	$222.17	$44.43	Released, 5 of 5 checks passed	Captures 2XX57477S7535341X and 76U34250RE4380745. Payouts DZNF5GE6AEUHN and Y4BT2Q3N8QR7J
deal-0002	Landing page copy	$324.21	$64.84	Released, 5 of 5 checks passed	Captures 5V6374204C639882V and 0FJ792206C854335V. Payouts A2EXK2WL4D4BG and L35MAUZEMNY92
deal-0003	Logo kit, weak delivery	$222.17	$44.43	Refunded in full, bond slashed, 0 of 5 checks passed	Refund 02S348440D980433R. Slash payout PWMV585KHNEM6
deal-0004	Cheat attempt	$180.96	$36.19	Settled graded after two rounds, 1 of 5 checks passed	Refund 9VH98649B6022315A. Slash payout X3BT5KNKEF8CL

rules; they do not prove the rules match human judgment on real work.

The live evidence is sandbox evidence. The four deals in Section IV-H moved test money between test accounts. Production use needs real seller onboarding under a Commerce Platform integration, which this build does not have. In this build both captures also land in the platform's own account, as Section III-D states.

The negotiation results come from rule-based agents with stated concession behavior. They show the mechanism terminates and respects the deal zone. They say nothing about how two language model negotiators would behave, and the multi-issue utility weights are simple by design.

The approval checkpoint needs a person. Each order waits for its account holder to approve it in PayPal's checkout. That is the right default for money movement, and it is also friction. Bounded pre-approval through mandates, the direction AP2 points in [9], is the open design problem I would take up next. The system also speaks one platform and one currency today: PayPal, in US dollars.

VII. CONCLUSION

Agent commerce has rails for authorization and settlement, and no layer that reads the work before the money moves. Pact adds that layer with two mechanisms that only make sense together: acceptance checks written by the negotiation that created the deal, and a per-deal seller bond that turns a failed delivery from a refund event into a loss for the side that caused it. Graded settlement keeps the outcome proportional, the contract hash ties the verdict to the money, and the ledger keeps the whole history open to inspection. The system ran end to end on the real PayPal sandbox, survived a 10,000 deal Monte Carlo run without creating or losing a cent, and caught a cheating seller twice on the same deal at a cost to the seller of most of its bond and 18 reputation points. The next steps are richer checks (schema checks on structured files, test-based checks on code), letting agents propose new check kinds instead of only haggling over values, and mandate-based pre-approval so repeat buyers stop approving every deal by hand.

AUTHOR NOTE

I designed Pact and I built it. The problem choice, the bond, the graded split, and the test batteries are mine. An AI assistant sometimes sped up the manual coding, with me controlling the code throughout. I ran every test myself and read every output. The defects the testing found are reported in Section IV-G alongside their fixes, not edited out.

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